

Kinematic Head Impact Detection

Machine-learning mini-project comparing HIKNet, RecursiveNet, support vector machine (SVM), and XGBoost for classifying kinematic recordings as real impacts or false positives.

Problem and dataset

The reference paper reports 527 mouthguard recordings collected from Stanford football athletes. Those original recordings are not available in this repository. Our experiment therefore uses synthetic data: 527 samples, six channels, 199 time points per channel, 264 impact labels, and 263 false-positive labels. Signals are simulated from damped pulses, higher-frequency components, and noise; they are not measured athlete data.

The generator seed for this comparison is 42; the holdout split seed is 0. The synthetic signals reproduce the stated format and class counts, not the real dataset distribution. These results therefore do not establish real-world performance.

Method and implementation

A stratified 70/30 split gives 369 training and 158 held-out evaluation samples (79 per class). HIKNet and RecursiveNet receive per-sample, per-channel standardized raw signals. A separate 15% of the outer training data is used for neural early stopping; the held-out set is scored only after training. SVM and XGBoost use the same 176 per-recording engineered features: peaks, durations, and spectral power features. SVM standardizes features in a training-fitted pipeline.

Model	Input	Main configuration
HIKNet	Standardized raw signals	150 filters; kernel 15; dropout 0.4; GAP head
RecursiveNet	Standardized raw signals	32/64 filters; 3x3 conv; dropout 0.25; dense 256
SVM	176 engineered features	RBF kernel; StandardScaler; default SVC parameters
XGBoost	Same 176 features	200 trees; depth 4; learning rate 0.05; subsampling 0.8

Neural networks use Adam (learning rate 0.001), batch size 32, binary cross-entropy with logits, at most 50 epochs, and early-stopping patience 5. XGBoost uses a binary logistic objective and log-loss fitting metric. XGBoost was selected as the additional model because boosted trees can learn nonlinear interactions in engineered tabular features.

Results, comparison, and limitations

Four-model results

Model	Accuracy	Precision	Specificity	Sensitivity	ROC-AUC	PR-AUC
HIKNet	0.911	0.901	0.899	0.924	0.946	0.933
RecursiveNet	0.848	0.857	0.861	0.835	0.917	0.910
SVM	0.791	0.871	0.899	0.684	0.905	0.890
XGBoost	0.892	0.897	0.899	0.886	0.933	0.892

On this single synthetic-data split, HIKNet had the highest accuracy; HIKNet had the highest precision; HIKNet, SVM, XGBoost had the highest specificity; HIKNet had the highest ROC-AUC; and HIKNet had the highest PR-AUC. These findings describe this split only and are not evidence that XGBoost outperforms the paper's models on Stanford recordings.

Paper result versus this experiment

The paper abstract reports HIKNet accuracy 98.2%, precision 97.6%, specificity 96.7%, and sensitivity 99.3%. Our HIKNet values are 91.1%, 90.1%, 89.9%, and 92.4%, respectively. This is a contextual comparison, not a like-for-like reproduction: the paper used real recordings, whereas this run used simulated signals and a different evaluation procedure.

Conclusions and limitations

The four-model pipeline runs on a common dataset split and reports accuracy, precision, specificity, sensitivity, ROC-AUC, and PR-AUC. The final comparison used synthetic generator seed 42 and outer split seed 0. The generator is artificial and encodes assumed class-related signal patterns. The result is one holdout run, not a multi-seed estimate and not validation on real mouthguard data.

The separate Wu 2017 workbook (387 training samples, 411 features, and a 32-sample test set) belongs to a different SVM workflow. It is not used for the SVM row above. The main experiment is implemented in `scripts/run_final_comparison.py`; metrics are saved to `results/final_comparison.csv` and `results/final_comparison.json`.